

Trym Sæther

Software Developer | M.Sc. Industrial Mathematics Student

saether.trym@gmail.com | +47 99 50 88 10 | [linkedin.com/in/trym-saether](https://www.linkedin.com/in/trym-saether) | github.com/trymsaether

Profile

Mathematically oriented software developer with a background in Industrial Mathematics, Electrical Engineering, and professional simulation software development at Infineon in Dresden. Interested in simulation software, numerical methods, scientific computing, EDA-related tools, compiler/tooling work, numerical optimization, and clear mathematical communication. Strong capacity for combining demanding studies, technical work, and volunteer responsibility.

Experience

Infineon Technologies

Dresden, Germany

Software Developer Intern – TITAN Simulation Software

Oct 2025 – Mar 2026

- Worked on TITAN, Infineon's in-house analog circuit simulation software, focusing on updates and improvements to the Verilog-A module.
- Developed in a large engineering codebase using primarily C++ and Fortran, with exposure to SPICE-like simulation workflows and model integration.
- Applied mathematical and engineering background to professional simulation software development in a semiconductor R&D environment.
- Balanced full-time M.Sc. studies at NTNU with full-time software development work abroad, demonstrating discipline, independence, and high workload capacity.

Studentersamfundet i Trondheim

Trondheim, Norway

Volunteer – Markedsføringsgjengen, Redaksjonen

Spring 2025 – Autumn 2026

- Volunteer in the editorial/social media section of Studentersamfundet's marketing group, working with a collaborative student team.
- Contribute to social media communication across accounts with approximately 100k followers in total.
- Help promote events and community activity for one of Trondheim's central student organizations while building strong teamwork experience and friendships.

Norges Håndballforbund

Norway

National Federation Referee

Jan 2013 – Present

- Referee at a high national level in Norwegian handball, requiring rapid decision-making, clear communication, and leadership under pressure.
- Developed judgment, conflict handling, presence, and responsibility in high-tempo environments.

Trondheim Kommune

Trondheim, Norway

Support Worker / Støttekontakt

Jan 2022 – Jan 2023

- Supported a young person with physical and psychological challenges through structured social, educational, and leisure activities.
- Built trust, responsibility, and communication skills through regular one-to-one support work.

Education

NTNU – Norwegian University of Science and Technology

Trondheim, Norway

M.Sc. / Civil Engineer (Sivilingeniør), Physics and Mathematics – Industrial Mathematics Expected Spring 2027

- Focus on applied mathematics, mathematical modeling, numerical analysis, probability, statistics, and modern information technology.
- Relevant academic interests include numerical methods, optimization, differential equations, stochastic modeling, and scientific computing.

- Background in electronics, programming, measurement/control systems, and engineering fundamentals.
- Connects physical systems, mathematical modeling, and software implementation.

Selected Technical Work

VAMSC – Early-stage C++20 compiler project for Verilog-AMS, focused on a frontend-first vams check-slice with source management, diagnostics, lexer/parser, AST construction, semantic analysis, and token/AST/sema output modes.

C++20 · Verilog-AMS · Compiler frontend · Diagnostics · CMake/CTest · Regression testing

SVM Classifier – Optimization and machine-learning project implementing soft-margin support vector machines from the primal and dual formulations, including a primal gradient solver, dual projected-gradient solver, support-vector recovery, and reusable kernels.

Python · NumPy · Numerical optimization · Machine learning · Linear/polynomial/RBF/Laplacian kernels

Numerical Image Processing – Numerical image-processing prototype for Poisson-style image blending: constructs curve-based masks, finite-difference Laplacian systems, boundary terms, and sparse linear solves to blend selected image regions channel by channel.

Python · NumPy/SciPy · Sparse linear algebra · Finite differences · Image processing

AFEM – Adaptive finite element / numerical methods project focused on computational approximation and scientific software.

Finite elements · Numerical PDEs · Scientific computing · Simulation

FortDAE – Software/tooling project related to differential-algebraic equation workflows and simulation.

DAEs · Simulation · Numerical methods · Tooling

Particle Transport with Ocean Currents – Scientific computing project modeling particle trajectories in ocean-current data.

Simulation · Numerical integration · Vector fields · Visualization

Course Notes & Book – Mathematical notes and book-style writing focused on clear technical explanation.

LaTeX · Mathematical communication · Numerical methods · Technical writing

Technical Skills

Languages	C++, Fortran, Python, C, JavaScript, TypeScript, Perl, Rust, LaTeX, Verilog-AMS/SPICE
Simulation / modeling	Verilog-AMS & SPICE, circuit simulation concepts, numerical methods, differential equations, scientific computing
Numerical / ML	optimization, support vector machines, sparse linear algebra, finite differences, image processing, visualization
Software	Git, CMake/CTest, large codebases, compiler frontends, diagnostics, regression testing, technical documentation
Communication	Mathematical notes, book-style writing, social media communication, editorial collaboration

Additional Experience

Norwegian Armed Forces

Medic

- Developed responsibility, composure, and practical judgment in demanding settings.

Posten AS

Logistics

Central Norway

Aug 2015 – Jul 2019

- Worked with logistics and distribution organization for advertising in Central Norway.